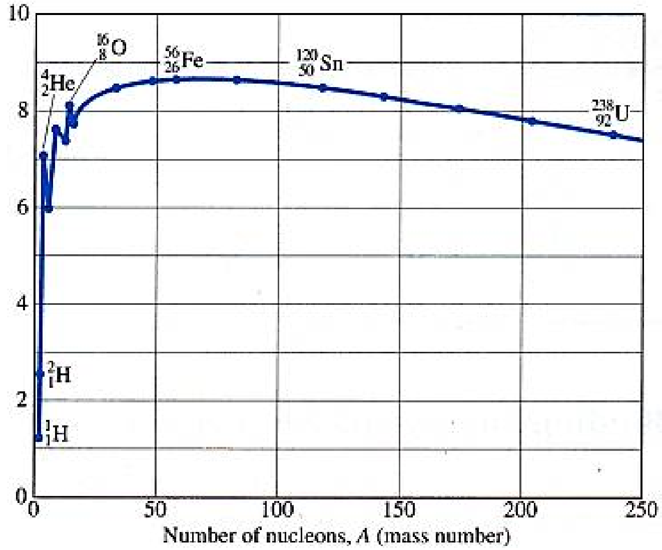
	1.8 s	
10(a)	The half-life $t_{\frac{1}{2}}$ of a radioactive nuclide is the average time taken for the nuclei to disintegrate to half its initial number of nuclei.	
10(b)(i)1.	Probability of decay per second is the decay constant λ. $\lambda = \dfrac{\ln 2}{t_{\frac{1}{2}}} = \dfrac{\ln 2}{45 \times 24 \times 60 \times 60} = 1.8 \times 10^{-7}\ \text{s}^{-1}$	
10(b)(i)2.	$A = \lambda N$ $\Rightarrow N = \dfrac{A}{\lambda} = \dfrac{8.5 \times 10^{7}}{1.8 \times 10^{-7}} = 4.8 \times 10^{14}$	
10(b)(i)3.	$A = A_o \exp(-\lambda t)$ $\Rightarrow \dfrac{A_o}{5} = A_o \exp\left(-\left(1.8 \times 10^{-7}\right) t\right)$ $t = 8.94 \times 10^{6}\ \text{s}$	

10(b)(ii)	The time is shorter as some of the ^{59}Fe is removed from the body through excretion.	
10(c)(i)		
10(c)(ii)	When a nuclei of lower mass number undergoes nuclear fusion, the products have higher mass number and thus have higher binding energy per nucleon. More energy is then released when they are formed from their component particles as compared to the energy that is needed to break the reactant into their constituent particles, resulting in a net release of energy.	
10(c)(iii) 1.	The nucleus *X* is a neutron.	
10(c)(iii) 2.	Change in mass $= \Delta m$ $=\left[(2.014102+3.016050)-(4.002603+1.008665)\right]u$ $=0.018884u = 3.13\times10^{-29}$ kg Energy produce by a single reaction $=\Delta mc^2$ $=0.018884\times(1.66\times10^{-27})\times(3.0\times10^{8})^2$ $= 2.82\times10^{-12}$ J Number of reactions possible from 150 kg of mixture $=\dfrac{150}{(2.014102+3.016050)u}$ $=1.796\times10^{28}$ Energy produced $=(1.796\times10^{28})(2.82\times10^{-12})$ $= 5.07\times10^{16}$ J	

10(c)(iii) 3.	Amount of energy available for the submarine $= 0.15\left(5.07\times10^{16}\right) = 7.61\times10^{15}$ J Length of time $= \dfrac{7.61\times10^{15}}{110\times10^{6}} = 6.91\times10^{7}$ s $= 2.2$ years	